Scalar k:2k effective boundary patches and directed-rounding seed

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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Purpose and scope

R-113 makes every analytic boundary patch in R-112 effective, proves a sharper global phase-minimum floor, and executes the first rigorous Arb certificate on a genuinely mixed three-parameter box. It does not cover the remaining central compact set and therefore does not prove the mixed all-q scalar theorem, $\mathrm{OVERLAP}_{\mathrm{src}}$**, Nelson, or Sector A.**
The historical R-085 formulas (4.10)–(4.11) and (6.5) are not reopened. R-088 removed the spurious outer 2^j weight, R-092/R-103 close the declared regular complete-packet and REG owners, and R-105 shows that visitwise Cartan and rational atoms do not descend under subdivision. The live frontier is the R-112 stationary scalar $k{:}2k$ normalizer, followed—only if that normalizer closes—by the full A1 one-fresh-root owner cluster and a separate one-use source/sextic aggregation.
This note advances that live scalar frontier in three ways. First, it extracts explicit rational projective wedges, eight explicit origin cones, and explicit covariance-face widths. Second, two exact completions improve the high-q phase floor, including a uniform zero-amplitude theorem for $\tau \geq 13$. Third, outward-rounded Arb sums prove one strict mixed box by two independent radial partitions. The unfilled central cover is recorded as open, not hidden behind the successful seed box.

Both executables exactly reconstruct every newly derived arithmetic constant assigned to their audit. Analytic inequalities not reducible to finite arithmetic are proved in the text; in particular, the face analysis below extends the R-111 argument, while the executables check its four substitutions and branch translations. The only decimal-like output is an Arb interval endpoint whose outward rounding is part of the proof. Preliminary Gauss–Laguerre scans are not evidence and are not used below.

1 R-112 compact target

Retain the R-112 coordinates

$$c = \frac{v}{v+4w}, \qquad s = 1-c, \qquad b = \frac{A^2}{v+4w}, \qquad \tau = q(v+4w)^2, \qquad x = b\tau. \tag{1.1}$$

For independent unit exponentials T, U, set

$$\rho = \frac{cT}{2}, \qquad \sigma = \frac{sU}{8}, \tag{1.2}$$

and

$$L = \rho + 4\sigma - \frac{1}{2}, \quad Q = \rho^2 + 10\rho\sigma + 4\sigma^2 - \rho - \sigma, \quad B = 6\rho\sqrt{\sigma}\cos\phi. \tag{1.3}$$

The normalized packet and deterministic covariance square are

$$F = bL + \sqrt{b}\,B + Q, \qquad K = b^2K_2 + bK_1 + K_0, \tag{1.4}$$

where

$$K_2 = \frac{c^2+s^2}{2}, \tag{1.5}$$

$$K_1 = c^3 + \frac{5}{2}c^2s + cs^2 + \frac{s^3}{4}, \tag{1.6}$$

$$K_0 = \frac{5c^4}{4} + \frac{c^3s}{4} + \frac{25c^2s^2}{16} + \frac{cs^3}{4} + \frac{5s^4}{64}. \tag{1.7}$$

The exact reserve is

$$\Delta = K - 2\operatorname{Var}(F) = \frac{c^4}{4} + \frac{c^2s^2}{4} + \frac{s^4}{64} \geq \frac{1}{100}. \tag{1.8}$$

With

$$M(c,x,\tau) = \mathbb{E}\left[e^{-xL-\tau Q} I_0(6\sqrt{x\tau}\,\rho\sqrt{\sigma})\right], \tag{1.9}$$

the target gap is

$$G(c,x,\tau) = \frac{x^2K_2 + x\tau K_1 + \tau^2K_0}{4} - \log M(c,x,\tau). \tag{1.10}$$

R-112 already proves the target outside

$$(2x+\tau)^2 \leq 32x + 20\tau, \qquad 0 \leq c \leq 1, \quad 0 \leq x \leq 8, \quad 0 \leq \tau \leq 20, \tag{1.11}$$

and supplies $D_0 \geq x^3/480$, $D_1 \geq 0$, and $D_2 \geq x^2\Delta/4$ in the projective expansion.

2 Effective projective wedges

For $x > 0$, put $\delta = \tau/x$ and $\varepsilon = \sqrt{\delta}$. Under the leading xL tilt, phase symmetry makes the logarithm even in ε. For $Y = \sqrt{U}$ use the uniform bounds

$$|B| \leq B_* = \frac{5}{12}TY, \tag{2.1}$$

$$|Q| \leq Q_* = \frac{T^2}{4} + \frac{5TY^2}{8} + \frac{Y^4}{16} + \frac{T}{2} + \frac{Y^2}{8}. \tag{2.2}$$

Indeed

$$\frac{4}{27} - c^2(1-c) = \frac{(3c-2)^2(3c+1)}{27}, \tag{2.3}$$

and the slightly enlarged rational coefficient 5/12 avoids any irrational moment oracle.

For $q_* \geq 2\varepsilon$ let $R_* = B_* + q_*Q_*$. The R-112 floor and $b = \varepsilon^{-2} \geq 10$ imply

$$e^{-x(L+\varepsilon B+\varepsilon^2Q)} \leq e^{x/2}. \tag{2.4}$$

For $p_k = \lceil k/2 \rceil$, differentiation and the exponential Bell polynomial give

$$|M^{(k)}(\varepsilon)| \leq A_k x^{p_k}, \tag{2.5}$$

where

$$A_k = E_X \sum_{j=0}^{\lfloor k/2 \rfloor} \frac{k!}{(k-2j)!\,j!} X^{k-j-p_k} \overline{\mathbb{E}}\Big[R_*^{k-2j} Q_*^j\Big]. \tag{2.6}$$

Here $x \le X$, $e^{x/2} \le E_X$, and the rational moment majorant is

$$\overline{\mathbb{E}}[T^a Y^{2n}] = a!n!, \tag{2.7}$$

$$\overline{\mathbb{E}}[T^a Y^{2n+1}] = 2a! \frac{(2n+2)!}{4^{n+1}(n+1)!}. \tag{2.8}$$

Equation (2.8) uses only $\sqrt{\pi} < 2$.

Jensen gives $M \ge 1$. Substitution of (2.5) into the exact sixth derivative of $\log M$ yields

$$|(\log M)^{(6)}(\varepsilon)| \le C_X x^3. \tag{2.9}$$

The independently reconstructed rational constants are

X	E_X	q_*	C_X
1	2	3/16	181292201822646474659735/273593677362757632
2	3	1/8	267774784673603525005/79164837199872
4	8	1/16	274770569726043833586365/1068725302198272
8	55	3/256	5748965910841593523942194993404275/373546567492618420224

(2.10)

They are respectively below

$$700000, \quad 3500000, \quad 260000000, \quad 16000000000000. \tag{2.11}$$

Taylor's theorem and evenness now give

$$G = D_0 + D_1\delta + D_2\delta^2 + \mathcal{R}_3, \qquad |\mathcal{R}_3| \le \frac{C_X}{720} x^3 \delta^3. \tag{2.12}$$

Exact integer comparisons with $D_0 \ge x^3/480$ prove

$$\boxed{G \ge \frac{x^3}{960} > 0} \tag{2.13}$$

on the four effective wedges

$$\begin{array}{c|c} 0 < x \le 1 & \tau/x \le 2^{-7} \\ 1 < x \le 2 & \tau/x \le 2^{-8} \\ 2 < x \le 4 & \tau/x \le 2^{-10} \\ 4 < x \le 8 & \tau/x \le 2^{-15}. \end{array} \tag{2.14}$$

3 Effective origin and covariance faces

On $s = 1 - c$ one has

$$K_2 \le \frac{1}{2}, \qquad K_1 \le 1, \qquad K_0 \le \frac{5}{4}. \tag{3.1}$$

Combining these with $\Delta \ge 1/100$ in the R-112 Bernstein–MGF condition proves the following. If $\tau > 0$, $x/\tau \le R$, and

$$\tau \le \theta(R) := \frac{48}{25(8R+5)(2R^2+4R+5)}, \tag{3.2}$$

then $G \ge 0$. The eight dyadic cone thresholds for $R = 2^j$ are

$$\frac{48}{3575}, \ \frac{16}{3675}, \ \frac{48}{49025}, \ \frac{16}{94875}, \ \frac{48}{1931825}, \ \frac{16}{4743675}, \tag{3.3}$$

$$\frac{48}{109255025}, \qquad \frac{16}{285418875}. \tag{3.4}$$

Together with the first wedge in (2.14), they cover the complete square

$$0 \le x, \tau \le \theta, \qquad \theta = \frac{16}{285418875}. \tag{3.5}$$

Here $\tau = 0, x > 0$ belongs to the projective edge, while $(x, \tau) = (0, 0)$ is the exact equality point.

The exact R-111 face theorem can also be made quantitative. At $c = 0$ use

$$m_0(x,\tau) = \begin{cases} 5\tau^2/1792, & 16x + 5\tau \le 16, \\ \tau/[3(\tau+64)], & 16x + 5\tau > 16, \end{cases} \tag{3.6}$$

and at $c = 1$ use

$$m_1(x,\tau) = \begin{cases} 5\tau^2/112, & 4x + 5\tau \le 4, \\ \tau/[3(\tau+16)], & 4x + 5\tau > 4. \end{cases} \tag{3.7}$$

Here is the additional reserve behind the large branches. For $\tau > 0$, in the universal one-frequency face put $h = \tau/16$, $\alpha = 4x/\tau$ for $c = 0$, and $h = \tau/4$, $\alpha = x/\tau$ for $c = 1$. With $\ell = 1 + 2\alpha h$, $X \sim \mathrm{Exp}(\ell)$, and $W = (X-1)^2 - 1$,

$$\log \mathbb{E}_\ell e^{-hW} = h + \log \mathbb{E}_\ell e^{-h(X-1)^2}.$$

On $A = \{X \le 1/2\}$, $(X-1)^2 \ge 1/4$, and

$$\Pr_\ell(A) = 1 - e^{-\ell/2} \ge \frac{\ell}{\ell+2} \ge \frac{1}{3}, \qquad 1 - e^{-h/4} \ge \frac{h}{h+4}.$$

Thus $\log \mathbb{E}_\ell e^{-hW} \le h - h/[3(h+4)]$. Writing $d(y) = y^2/2 - y + \log(1+y) \ge 0$, the R-111 tilted identity gives, whenever $(4\alpha+5)h \ge 1$,

$$G_{\mathrm{face}} \ge d(2\alpha h) + h((4\alpha+5)h - 1) + \frac{h}{3(h+4)} \ge \frac{h}{3(h+4)}.$$

We used $1 - e^{-y} \ge y/(1+y)$ and $\log(1-z) \le -z$. On $(4\alpha+5)h \le 1$, the R-111 quadratic estimate instead leaves $5h^2/7$. Equality is assigned to this small branch in (3.6)–(3.7), although the large estimate is also valid there. These two universal margins give exactly the four displayed substitutions.

For the interior derivative, $I_1(z) \le z I_0(z)/2$ coefficientwise and the R-112 phase floor give

$$|\partial_c G| \le \Lambda(x,\tau), \tag{3.8}$$

$$\Lambda = 60000\left(x + \frac{5\tau}{2} + \frac{9x\tau}{16}\right) + \frac{x^2}{4} + \frac{x\tau}{4} + \frac{19\tau^2}{16}. \tag{3.9}$$

Therefore, whenever $\Lambda > 0$,

$$c \le \frac{m_0}{2\Lambda} \implies G \ge \frac{m_0}{2}, \qquad 1 - c \le \frac{m_1}{2\Lambda} \implies G \ge \frac{m_1}{2}. \tag{3.10}$$

After removing (2.14), (3.2), and (3.10), every unresolved point lies in

$$\boxed{\kappa \le c \le 1-\kappa, \quad 0 \le x \le 8, \quad \theta \le \tau \le 20, \quad (2x+\tau)^2 \le 32x + 20\tau,} \tag{3.11}$$

where

$$\kappa = \frac{1}{100552401097327200}. \tag{3.12}$$

This very conservative width separates the residual problem from both covariance faces and from $\tau = 0$, and leaves a rational compact set. It does not remove the zero-amplitude boundary: the segment $x = 0$, $\theta \le \tau < 13$ remains unresolved. At radial cutoff 80, the uniform tail is below

$$120000(2/5)^{80} < 2 \times 10^{-27}. \tag{3.13}$$

4 Sharper phase-minimum floors

Minimize the phase by taking $B = -6\rho\sqrt{\sigma}$. Two independent exact completions are

$$\begin{aligned} F_{-} &= (\rho + 2\sigma)^2 - \left(1 + \frac{b}{2}\right)(\rho + 2\sigma) + (5b + 1)\sigma \\ &\quad + 6\rho\left(\sqrt{\sigma} - \frac{\sqrt{b}}{2}\right)^2 - \frac{b}{2}, \end{aligned} \tag{4.1}$$

and

$$\begin{aligned} F_{-} &= \rho^2 + (b/10 - 1)\rho + 4\sigma^2 + (4b - 1)\sigma \\ &\quad + 10\rho\left(\sqrt{\sigma} - \frac{3\sqrt{b}}{10}\right)^2 - \frac{b}{2}. \end{aligned} \tag{4.2}$$

Consequently

$$F \geq -B_6(b), \qquad B_6(b) = \frac{1}{4} + \frac{3b}{4} + \frac{b^2}{16}, \tag{4.3}$$

and

$$F \geq -B_{10}(b), \quad B_{10}(b) = \frac{b}{2} + \frac{(1 - b/10)_+^2}{4} + \frac{(1 - 4b)_+^2}{16}. \tag{4.4}$$

Thus, with $B_{\text{new}} = \min(B_6, B_{10})$, the scalar target holds whenever

$$\tau \geq \frac{4B_{\text{new}}(b)}{b^2 K_2 + bK_1 + K_0}. \tag{4.5}$$

At $b = 0$, $Q \geq -1/4$ is sharp. Moreover

$$K_0(c) = \frac{153c^4 - 156c^3 + 82c^2 - 4c + 5}{64} \geq \frac{1}{13}. \tag{4.6}$$

The exact Sturm check uses

$$1989c^4 - 2028c^3 + 1066c^2 - 52c + 1, \tag{4.7}$$

which has no real roots and is positive at zero. Therefore the entire zero-amplitude covariance simplex is proved for

$$\boxed{b = 0, \qquad \tau \geq 13.} \tag{4.8}$$

5 First directed-rounding mixed box

Let

$$\mathcal{R} = \frac{x^2 K_2 + x\tau K_1 + \tau^2 K_0}{4}, \qquad z = 6\sqrt{x\tau}\rho\sqrt{\sigma}. \tag{5.1}$$

The exact sum-of-squares coordinate

$$\begin{aligned} D &= \tau(\rho - 1/2)^2 + 4\tau(\sigma - 1/8)^2 + 4x\sigma \\ &\quad + \rho\{(\sqrt{x} - 3\sqrt{\tau\sigma})^2 + \tau\sigma\} - \frac{x}{2} - \frac{5\tau}{16} \end{aligned} \tag{5.2}$$

satisfies $D = xL + \tau Q - z$. With $\mathrm{i0e}(z) = e^{-z} I_0(z)$,

$$e^{-\mathcal{R}} M = \int_0^\infty \int_0^\infty e^{-T-U-D-\mathcal{R}}\, \mathrm{i0e}(z)\, dT\, dU. \tag{5.3}$$

Since i0e decreases on $[0,\infty)$, a parameter box P and radial rectangle C have the rigorous upper contribution

$$|C| \exp\left(\sup_{P\times C}[-T-U-D-\mathcal{R}]\right) \mathrm{i0e}\left(\inf_{P\times C} z\right). \tag{5.4}$$

Every operation in (5.4) is an Arb ball operation with outward rounding. In particular, every square-root range is the Arb union of its two outward- rounded endpoint balls; both executables test a width-2^{-200} interval and an interval starting at zero. The floating heap key changes only the order of valid dyadic subdivisions.

The primary partition uses 20000 cells and proves the complete box

$$(c,x,\tau) \in [49/100, 51/100] \times [99/100, 101/100]^2. \tag{5.5}$$

Its normalized total upper endpoint is

$$0.96684937293277445705287215953756468063 84 < 1. \tag{5.6}$$

The independent implementation uses a different initial partition, refinement priority, and 30000 cells. It obtains

$$0.97600139632813314867472523544722302486 58 < 1. \tag{5.7}$$

Both enclose the radius-50 exterior analytically and verify that the whole parameter box lies strictly inside (1.11). Thus (5.5) is the first certified genuinely mixed strict-core box. It is a seed for a global cover, not that cover itself.

6 Failed contraction and remaining theorem

The exact cross contraction

$$e^{-x\rho-10\tau\rho\sigma} I_0(6\sqrt{x\tau}\rho\sqrt{\sigma}) \le e^{-\tau\rho\sigma} \tag{6.1}$$

follows from $I_0(z) \le e^z$ and $2\sqrt{ab} \le a+b$. It reduces the Bessel integral to a positive quadratic radial integral and then to one stable erfcx integral. However, the contracted surrogate has

$$\mathbb{E}\widetilde{F}_b = -\frac{c(8b+9s)}{16} < 0 \tag{6.2}$$

in the mixed interior. Its normalized log-Laplace transform therefore has an artificial positive $O(\tau)$ origin debt, whereas the target begins at $O(\tau^2)$. A proof that globally replaces the original Bessel factor by this contracted surrogate cannot close the origin without an additional mean-debt cancellation. The failure is registered as

$$\texttt{NG-2026-07-28-A13-K2K-BESSEL-CROSS-CONTRACTION-ORIGIN-DEBT}. \tag{6.3}$$

It is not a target counterexample.

The remaining scalar decision problem is a fail-closed directed-rounding branch-and-bound on (3.11) after subtracting the certified box (5.5) and the analytic high-q region (4.5). A proof requires a finite accepted cover; any unresolved or nonpositive box must be retained rather than inferred positive. Even a successful scalar closure would not identify the full A1 one-fresh-root owner cluster or prove the separate source/sextic one-use ledger. R-103 REG and R-087 fixed-cutoff CORE retain their accepted scopes.

7 Devil's-advocate review

1. **Objection:** A positive numerical sample was promoted to a theorem. **Verdict: DISMISSED.** Equations (5.3)–(5.4) are upper Riemann sums evaluated by directed-rounding Arb balls; both programs assert an upper endpoint strictly below one. Ordinary quadrature scans are absent from the evidence contract.

2. **Objection:** The successful box proves the entire mixed scalar normalizer. **Verdict: UPHELD AS A FORBIDDEN READING.** Only (5.5) is certified. The manifest, status card, footer, and integrated verifier all require the global compact-cover flag to remain false.

3. **Objection:** The projective Taylor remainder silently assumes irrational half-moments or a sign for every coefficient. **Verdict: DISMISSED.** Odd half-moments are enlarged using $\sqrt{\pi} < 2$, the sixth logarithmic derivative is bounded by a rational Bell table, and only the already proved D_0, D_1, D_2 signs are used. R-112's negative D_3 fixture remains valid.

4. **Objection:** The face widths vanish at the equality origin, so the stated central set is not compact. **Verdict: DISMISSED.** The eight origin cones and effective projective wedge first impose $\tau \geq \theta > 0$ on every unresolved point; only then is the explicit c-width floor $\kappa > 0$ taken.

5. **Objection:** The sharper floor closes Sector A. **Verdict: UPHELD AS A FORBIDDEN READING.** It closes only the explicit scalar high-q subregion. Full A1 embedding, one-use aggregation, $\text{OVERLAP}_{\text{src}}$, Nelson, removals, the interacting measure, and Sector A remain open.

6. **Objection:** The old R-085 Cartan and rational atoms are now proved by the scalar calculation. **Verdict: UPHELD AS A FORBIDDEN READING.** Those historical weighted/labelled-owner atoms are non-load-bearing or representation dependent under R-088/R-105. This result neither proves nor needs their literal historical forms.

8 Result footer

```
Result ID: A13-CLASSII-SCALAR-K2K-EFFECTIVE-BOUNDARY-
           DIRECTED-ROUNDING-SEED
Precise statement: R-113 makes the R-112 projective, origin, and
  covariance-face patches explicit; proves sharper phase floors including
  b=0, tau>=13; and certifies one strict mixed Arb box.  The global mixed
  compact cover remains open.
Scope: Stationary scalar k:2k model only; finite compact reduction and one
  certified mixed box, not a global mixed all-q theorem or full A1 embedding.
Dependencies: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION;
  R-103; R-105; R-111; R-112.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command:
  E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/
  a13_classii_scalar_k2k_effective_boundary_interval_seed_verify.py
Expected output: Integrated R-113 PASS; primary 75/75 and independent 64/64;
  both Arb normalized upper endpoints are below one; global-cover flag false.
Falsification gate: Any failed exact identity, rational threshold, Sturm
  check, Arb upper endpoint, source/PDF hash, or a claim that the remaining
  central cover is complete.
Tier before / after: T4 / T4.
No-overclaim statement: No mixed all-q scalar theorem, full A1 owner cluster,
  one-use source/sextic aggregation, OVERLAP_src, Nelson bound, removal limit,
  interacting measure, Sector-A closure, or higher tier is established.
Next required action: Run a fail-closed directed-rounding branch-and-bound
  on the residual set (3.11), accepting scalar closure only after a finite
  certified cover; then attempt owner-preserving full-A1 embedding and the
  separate one-use aggregation.
```